This assignment will not be graded and is only for practice.

3.1 Level 1:

Consider a system of linear equations (System 1):

$$\begin{aligned} -2x_1 + 3x_2 + x_3 &= 1 \\ -x_1 + x_3 &= 0 \\ 2x_2 &= 5 \end{aligned}$$

Answer questions 1 and 2 based on the above data.

1) **1 point**

If the matrix representation of system (1) is $Ax = b$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, then

- [x] $A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$
- [ ] $b = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$
- [ ] $A = \begin{bmatrix} -2 & -1 & 0 \\ 3 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$
- [x] $A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$

Yes, the answer is correct.

Score: 1

Targeted Feedback:

Feedback:

Accepted Answers:

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$

$A = \begin{bmatrix} -2 & 3 & 1 \\ -1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix}$

2) System (1) has **1 point**
[Hint: Solve for x_1, x_2, and x_3.]

- (selected) a unique solution.
- no solution.
- infinitely many solutions.
- None of the above.

Yes, the answer is correct.
Score: 1

Accepted Answers:

a unique solution.

3) Choose the set of correct options. **1 point**
[Hint: Think of A as a 2×2 or 3×3 matrix, and b accordingly.]

- [x] Every system of linear equations has either a unique solution, no solution or infinitely many solutions.
- [ ] If each equation of a system of linear equations is multiplied by a non-zero constant c, then the solution of the new system of equations is c times the solution of the old system of equations.
- [x] If $Ax = b$ is a system of linear equations which has a solution, then the system of linear equations $cAx = b$, where $c = 0$, will also have a solution.
- [x] If $Ax = b$ is a system of linear equations which has a solution, then $\frac{1}{c}Ax = b$, where $c = 0$, will also have a solution.

Yes, the answer is correct.
Score: 1

Accepted Answers:

Every system of linear equations has either a unique solution, no solution or infinitely many solutions.
If $Ax = b$ is a system of linear equations which has a solution, then the system of linear equations $cAx = b$, where $c = 0$, will also have a solution.
If $Ax = b$ is a system of linear equations which has a solution, then $\frac{1}{c}Ax = b$, where $c = 0$, will also have a solution.

4) The Plane 1 and Plane 2 in Figure M2W1AQ3, correspond to two different linear equations, which form a system of linear equations. ***1 point***

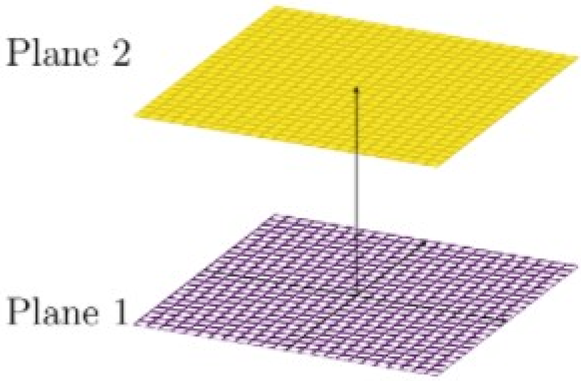

Figure M2W1AQ3

The above system of linear equations has
[Hint: If a system of linear equations has a solution, then the point corresponding to the solution must lie on each plane corresponding to each linear equation of the given system.]

- ○ a unique solution.
- ◉ no solution.
- ○ infinitely many solutions.
- ○ None of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:

no solution.

5) Consider the geometric representations (Figures (a), (b), and (c)) of three systems of linear equations. ***1 point***

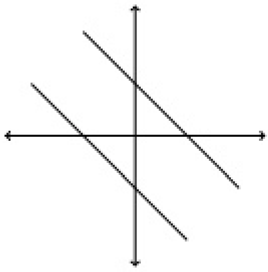

Figure (a)

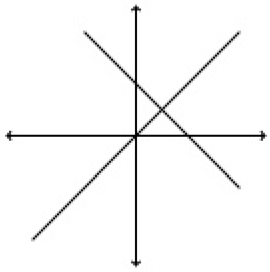

Figure (b)

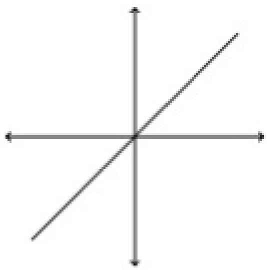

Figure (c)

Choose the set of correct options.

- [x] Figure (a) represents a system of linear equations which has no solution.
- [ ] Figure (a) represents a system of linear equations which has infinitely many solutions.
- [x] Figure (b) represents a system of linear equations which has a unique solution.
- [ ] Figure (b) represents a system of linear equations which has infinitely many solutions.
- [x] Figure (c) represents a system of linear equations which has infinitely many solutions.
- [ ] Figure (c) represents a system of linear equations which has no solution.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Figure (a) represents a system of linear equations which has no solution.
Figure (b) represents a system of linear equations which has a unique solution.
Figure (c) represents a system of linear equations which has infinitely many solutions.

3.2 Level 2:

6) Consider a system of equations: $\begin{aligned} 2x_1 + 3x_2 &= 6 \\ -2x_1 + kx_2 &= d \\ 4x_1 + 6x_2 &= 12 \end{aligned}$ Choose the set of correct options. **1 point**

- [ ] $Ax = b$ represents the above system, where $x = [\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}], A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$, and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$
- [ ] The system has no solution if $k = -3, d = 0$.
- [ ] The system has a unique solution if $k = 3, d = 0$.
- [ ] The system has infinitely many solutions if $k = -3, d = 6$.
- [ ] The system has infinitely many solutions if $k = -3, d = -6$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$Ax = b$ represents the above system, where $x = [\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}], A = \begin{bmatrix} 2 & 3 \\ -2 & k \\ 4 & 6 \end{bmatrix}$, and $b = \begin{bmatrix} 6 \\ d \\ 12 \end{bmatrix}$
The system has no solution if $k = -3, d = 0$.
The system has a unique solution if $k = 3, d = 0$.
The system has infinitely many solutions if $k = -3, d = -6$.

7) Let x_1 and x_2 be solutions of the system of linear equations $Ax = b$. Which of the following options are correct? **1 point**

- [ ] $x_1 + x_2$ is a solution of the system of linear equations $Ax = b$.
- [ ] $x_1 + x_2$ is a solution of the system of linear equations $Ax = 2b$.
- [ ] $x_1 - x_2$ is a solution of the system of linear equations $Ax = b$.
- [ ] $x_1 - x_2$ is a solution of the system of linear equations $Ax = 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x_1 + x_2$ is a solution of the system of linear equations $Ax = 2b$.
$x_1 - x_2$ is a solution of the system of linear equations $Ax = 0$.

8) Let v be a solution of the systems of linear equations $A_1 x = b$ and $A_2 x = b$. Which of the following options are correct ? **1 point**

- [ ] v is a solution of the system of linear equations $(A_1 + A_2)x = b$.
- [ ] v is a solution of the system of linear equations $(A_1 + A_2)x = 2b$.
- [ ] v is a solution of the system of linear equations $(A_1 - A_2)x = 0$.
- [ ] v is a solution of the system of linear equations $(A_1 - A_2)x = b$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

v is a solution of the system of linear equations $(A_1 + A_2)x = 2b$.
v is a solution of the system of linear equations $(A_1 - A_2)x = 0$.

9) Consider a system of equations: $\begin{matrix} x_1 - 3x_2 = 4 \\ 3x_1 + kx_2 = -12 \end{matrix}$ Where $k \in \mathbb{R}$. If the given system has a unique solution, then k should not be equal to

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -9

1 point

10) Consider two system of linear equations:

System 1:

$$x_1 + 2x_2 = -5$$
$$0x_1 - x_2 = 5$$

System 2:

$$-2x_3 + x_4 = -5$$
$$3x_3 + x_4 = 5$$

Suppose there is another system of linear equation given by

$$(1-2)(x_1 + x_3) + (2+1)(x_2 + x_4) = m$$
$$(0+3)(x_1 + x_3) + (-1+1)(x_2 + x_4) = n$$

for some real values of m and n. Find the value of $n - m$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 46

1 point

Your score is: 4/10